6 A capacitor of capacitance $C = 220\ \mu$F is charged by a battery which has an e.m.f. of 6.0 V.

(a) Calculate the charge on the capacitor when there is a potential difference of 6.0 V across it.

Charge = ... C [2]

(b) Calculate the energy stored in the capacitor when the potential difference across it is 6.0 V.

Energy stored = ... J [2]

(c) The capacitor and battery are used in a camera flashgun. The capacitor is first charged and then discharged through a lamp to produce a flash when a photograph is taken. It is then recharged from the battery through a circuit of total resistance *R*. The voltage *V* across the capacitor rises in the way shown in Fig. 6.1.

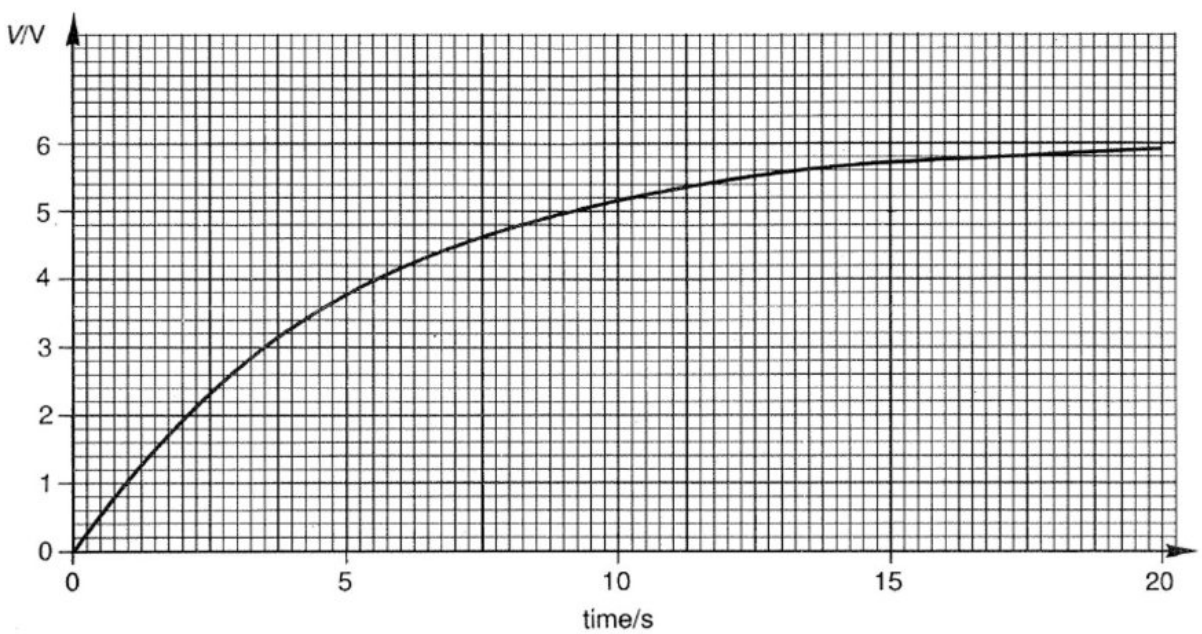


Fig. 6.1

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The graph shows the variation of the potential difference V across the capacitor with time during charging.
If the capacitor continued to charge at its initial rate, it would be fully charged after a time equal to the time constant CR of the circuit.

(i) On Fig. 6.1, draw the tangent to the graph at the origin. [1]

(ii) Use your tangent to find the time constant of the charging circuit.

time constant = ... s [2]

(iii) Deduce the value of R.

R = ... Ω [2]

[Total: 9]

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